

Pellagra

Etiology and Pathogenesis

5-Fluorouracil inhibits the conversion of tryptophan to niacin, and 6-mercaptopurine inhibits NAD phosphorylase, thereby reducing NAD production. Other medications associated with pellagra include phenytoin, chloramphenicol, azathioprine, sulfonamides, and antidepressants.

Clinical Findings

Pellagra is classically characterized by the "four Ds": **dermatitis**, **diarrhea**, **dementia**, and **death**.

The dermatitis is photosensitive and begins as erythematous, painful, or pruritic patches. The skin becomes edematous, and within several days may develop vesicles and bullae. These can rupture, leading to crusted erosions, or evolve into brown scales. Over time, the skin thickens into sharply demarcated, keratotic, hyperpigmented plaques. Painful fissures may form on the palms and soles, resembling goose skin.

The dorsum of the hands is the most commonly affected site. When the rash extends proximally, it typically involves the radial more than the ulnar side, forming the characteristic "gauntlet" of pellagra. On the face, a butterfly distribution may be seen, extending from the nose to the cheeks, chin, and lips. Involvement of the upper central chest and neck produces "Casal's necklace," which may extend down over the sternum as a "cravat."

Mucous membrane involvement includes **cheilitis**, **angular stomatitis**, **glossitis**, and ulcerations of the buccal mucosa and vulva.

Gastrointestinal symptoms—such as diarrhea, nausea, vomiting, abdominal pain, and anorexia—may be the earliest signs. Neurologic manifestations include insomnia, fatigue, nervousness, apathy, impaired memory, and

depression, which can progress to psychosis and dementia in advanced stages. Untreated pellagra results in death from multiorgan failure.

Laboratory Testing

Diagnosis is primarily clinical and supported by a rapid response to vitamin supplementation. Measurement of urinary metabolites—**N-methylnicotinamide** and **pyridone**—can aid in confirmation.

Treatment

The recommended daily intake of niacin is 15 to 20 mg, or approximately 60 mg of exogenous tryptophan. Treatment consists of 500 mg per day of **nicotinamide** or **nicotinic acid** administered over several weeks.

Nicotinamide is preferred due to the lower incidence of side effects; nicotinic acid commonly causes flushing and headache. Neuropsychiatric symptoms may improve within 24 to 48 hours, but skin lesions typically take 3 to 4 weeks to resolve.

Vitamin B₆ (Pyridoxine)

Etiology and Pathogenesis

Vitamin **B₆** refers to three interconvertible forms: **pyridoxine**, **pyridoxamine**, and **pyridoxal**. Humans cannot synthesize these compounds and must obtain them from the diet. Rich sources include meats, whole grains, vegetables, and nuts.

Food processing can reduce the bioavailability of vitamin B₆, which is absorbed by passive diffusion in the jejunum. The primary active form in the body is **pyridoxal-5-phosphate (PLP)**.